Hard Negative Mining for Knowledge Graph Link Prediction: A Controlled Ablation on FB15k-237

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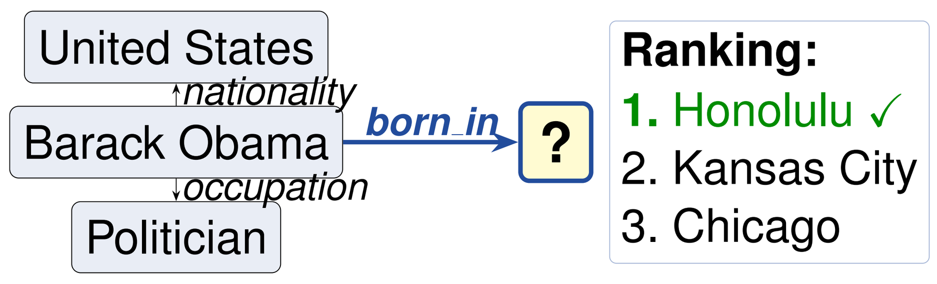
Background: Link Prediction

A **knowledge graph** (KG) is a directed multi-relational graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ where each triple (h, r, t) encodes a binary fact. **Link prediction** asks: given $(h, r, ?)$, rank all $|\mathcal{E}|$ candidate entities by plausibility. The standard metric is **MRR** (mean reciprocal rank of the correct entity in the filtered ranking).



Training requires *negative* triples (false facts). **Random** Bernoulli corruption yields easy negatives after a few epochs — gradient signal collapses. **Hard negatives** (triples the model still scores highly) carry more information but risk instability if used alone.

Dataset: FB15k-237

Property	Value
Entities	14,505
Relations	237
Train triples	272,115
Val / Test triples	17,526 / 20,438
Mean out-degree	19.7
Relation freq. range	37 – 15,989

Relation frequencies span **three orders of magnitude**, motivating Bernoulli side-choice and stratified slice evaluation across rare, medium, and frequent relation buckets.

Models & Training Setup

TransE: $f(h, r, t) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|_2$ (translation in $\mathbb{R}^d$)

RotatE: $f(h, r, t) = -\|\mathbf{h} \circ \mathbf{r} - \mathbf{t}\|$, $|r_k|=1$ (rotation in $\mathbb{C}^d$)

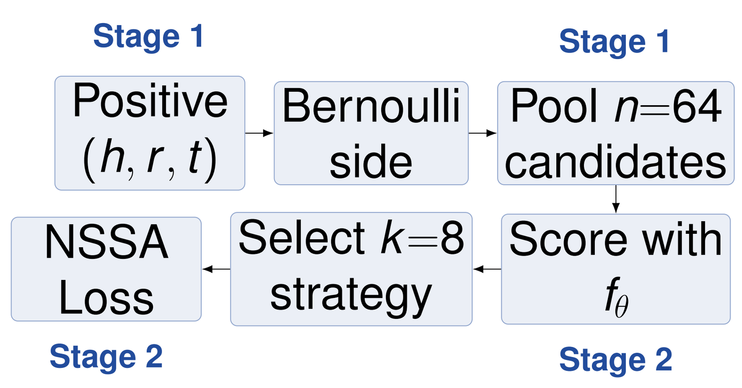
NSSALoss (self-adversarial):

$$\mathcal{L} = -\log \sigma(\gamma - f_+) - \sum_i p_i \log \sigma(f_i' - \gamma)$$

where $p_i = e^{\alpha f_i'} / \sum_j e^{\alpha f_j'}$ (margin γ=9.0, temp. α=1.0). Hard negatives get higher weight p_i, amplifying gradient signal.

All runs: $d = 128$, batch 1024, Adam lr$= 10^{-3}$, 50 epochs (patience 10), seed 42, $n = 64$ pool, $k = 8$ negs/positive.

Two-Stage Negative Sampling

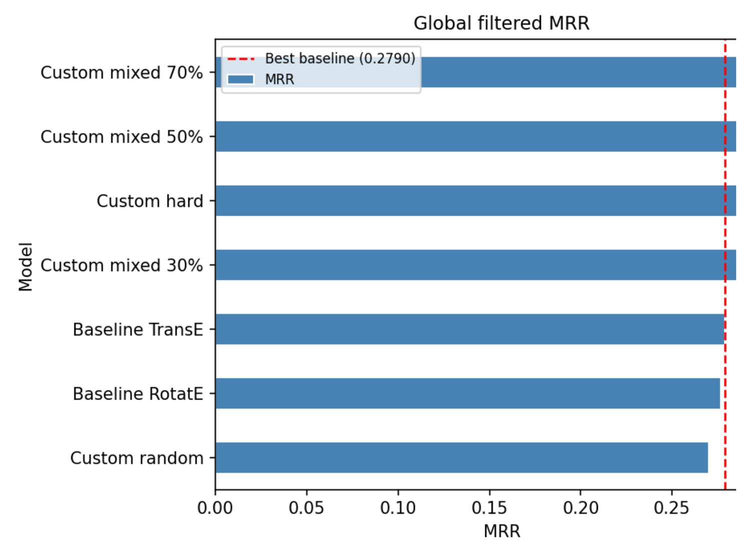


Stage 1: Bernoulli corruption draws n=64 candidates, filtering known-true triples.
Stage 2: selects k=8 via **Random** (control), **Hard** (top-k by f_θ), or **Mixed** p ($p \in \{0.3, 0.5, 0.7\}$). All strategies use the same pool — **zero extra compute cost**.

Conclusions

- **Hard negatives +10.4% MRR** over random sampling, consistent across all metrics and reproducible with seed 42.
- **Mixed 30% recovers 95% of the gain** at zero extra compute; recommended default for practitioners.
- **Gains concentrate on rare relations** (+3.6 pp) and high-degree tail entities, where hard negatives maximise

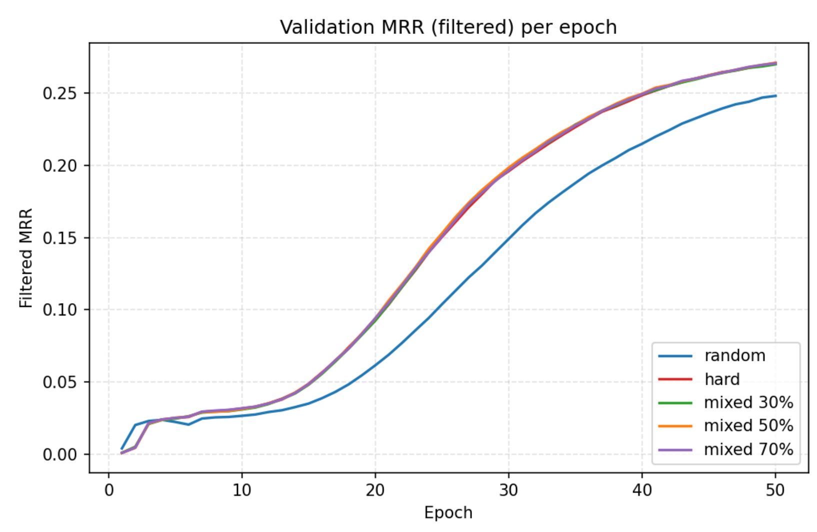
Main Results: Hard Negatives +10.4% MRR



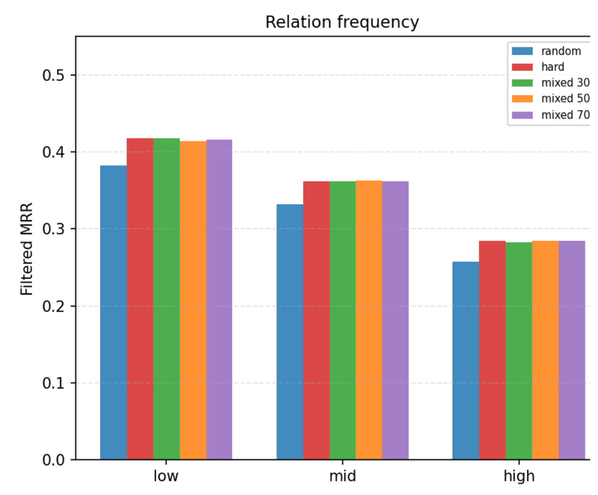
Strategy	MRR	H@1	H@3	H@10
Random (control)	0.270	0.189	0.300	0.428
Hard	0.298	**0.217**	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	**0.328**	0.460
Mixed 70%	**0.299**	**0.217**	**0.328**	**0.461**
Δ hard − random	+0.028	+0.028	+0.026	+0.031

All runs trained to the full 50-epoch budget. **Mixed 30%** recovers **95%** of the pure-hard gain ($\Delta_{\text{mix30}} = 0.0266$ vs. $\Delta_{\text{hard}} = 0.0280$).

Learning Dynamics & Slice Evaluation



Hard/mixed strategies **separate from random by epoch 15** and converge to a 7–8% higher plateau with no instability.



Rare relations gain most (+3.6 pp). Hard/Mixed dominate random in *every* slice: relation frequency, head out-degree, tail in-degree.

Strategy	Rare (+3.6 pp)	Medium (+3.0 pp)	Frequent (+2.8 pp)
Random	0.382	0.332	0.257
Hard	**0.418**	**0.362**	0.284
Mixed 70%	0.416	0.362	**0.285**

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